



National University of Computer & Emerging Sciences, Karachi
Fall-2021 FAST School of Computing



Final Examination

January 3rd 2021 Monday 9:00 am - 12:00 noon

Course Code: MT – 1003	Course Name: Calculus and Analytical Geometry
Instructor Name: Dr. Khusro Mian/ Dr. Fahad Riaz/ Ms. Asma Masood/ Mr. Nadeem Khan/ Ms. Afreen Naz/ Ms. Urooj/ Ms. Alishba Tariq/ Ms. Javeria Iftikhar/ Ms. Sadia Khan	
Student Roll No:	Section No:

Instructions:

- Return the question paper.
- **Attempt all questions. There are 10 Questions and 02 pages.**
- **Solve the paper according to the sequence given in the question paper.**
- Graphical Calculator is not allowed.

Time: 180 minutes

Max Marks: 100

Question 01:

03+03+04=10

✓(a) Show that the following function $f(x)$ is continuous on $(-\infty, +\infty)$.

$$f(x) = \begin{cases} 2x + 3 & x \leq 4 \\ 7 + \frac{16}{x} & x > 4 \end{cases}$$

✓(b) For the following $f(x) = \begin{cases} |x-2| & x > 0 \\ x+2 & x \leq 0 \end{cases}$ evaluate $\int_{-4}^6 f(x) dx$

✓(c) Find absolute extrema for $f(x) = 2x^3 + 3x^2 - 12x$ on $[-3, 2]$

Question 02:

05+05=10

Find the derivative of the function. [Any Two]

a) $f(x) = \cos(\sqrt{\sin(\tan \pi x)})$

✓(b) $f(x) = \frac{\tan 3x}{(x+7)^4}$

✓(c) $f(x) = [1 + \sin^3(x^5)]^{12}$

Question 03:

05+05=10

For the functions $x = 2 - y^2$ and $x = y^2$

✓(a) Sketch the region and find the area enclosed between two curves

✓(b) Find the volume of the solid obtained by revolving the curves along y axis.

Question 04:

05+05=10

For the following functions, does the integral diverges or converges?

(a) $\int_{-\infty}^{+\infty} (1-x)e^{-x} dx$

(b) $\int_0^1 \frac{1}{\sqrt{x}(x+1)} dx$

Question 05:

05+05=10

For the following functions, integrate using the method of substitution. (Attempt any two)

✓ $\int \frac{1}{\sqrt{2x-x^2}} dx$

✗ $\int_5^{10} \frac{\sqrt{x^2-25}}{x} dx$

✓ $\int \frac{1}{1-\sin x + \cos x} dx$

Question 06:

06+04=10

For the curve $x^2y + 3xy^3 - x = 3$ ✗ Find $\frac{dy}{dx}$ using implicit differentiation.

(b) Find an equation to the tangent line of the curve at (1,0).

Question 07:

05+05=10

Evaluate limits:

✓ (a) $\lim_{x \rightarrow +\infty} \left[\cos\left(\frac{2}{x}\right) \right]^{x^2}$

(b) $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + x} - x)$

Question 08:

03+03+04=10

Integrate the following:

(a) $\int e^{-3x} \sin 3x dx$

(b) $\int \frac{x^5 + x^2 + 2}{x^3 - x} dx$

✓ $\int \sin^5 x dx$

Question 09:

05+05=10

Show that the lines are skew and find the distance between them.

$L_1: x - 1 = 2t, y = 3t, z + 2 = -5t$

$L_2: x = t, y = 4 + t, z = 3t$

Question 10:

05+05=10

(a) Find the point of intersection between two lines.

$L_1: x + 1 = 4t, y - 3 = t, z - 1 = 0$

$L_2: x + 13 = 12t, y - 1 = 6t, z - 2 = 3t$

(b) Find the angles of intersection and the line of intersection between two planes.

$P_1: 2x - 4y + 4z = 6$

$P_2: 6x + 2y - 3z = 4$

The End ☺